Design and Computational Analysis of a Tesla Turbine

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Abstract- A Tesla turbine also called as a bladeless turbine in which fluid flows spirally on discs . It uses the Boundary layer phenomena & adhesion of fluid. This turbine stands out for its remarkable ability to achieve notably higher RPMs compared to conventional counterparts. This unique strength stems from its innovative design, where faster inlet fluid speeds result in heightened centrifugal force, driving increased rotation on the discs. This accelerated rotation not only enhances viscosity effects but also prolongs surface contact time between the fluid and discs. Consequently, more energy is transferred to the discs, further amplifying the turbine's efficiency and solidifying its position as a formidable player in energy generation.. Our project's main purpose is to study the performance of a Tesla turbine using various parameters like pressure and velocity , and by using air as a working fluid.

Keywords- Tesla Turbine, design, CFD Analysis, 3d printing, Viscosity, Blade less Turbine.

I. Introduction

The Tesla turbine's capability to achieve high rotational speeds is a notable feature attributed to its design and operating principle. Unlike conventional turbines which rely on blades or vanes to capture energy from fluid flow, the Tesla turbine utilizes a boundary layer effect to transfer momentum directly to its closely spaced disks. This design allows for minimal resistance to rotation, enabling the turbine to reach exceptionally high RPMs.

One key factor contributing to the high RPM capability is the absence of physical contact between moving parts, such as blades and casings, which minimizes frictional losses. The fluid flows smoothly over the disks, generating a uniform boundary layer that efficiently transfers momentum to the disks without inducing turbulence or drag

Furthermore, the compact and symmetrical design of the Tesla turbine facilitates balanced rotation, reducing vibration and mechanical stresses that could limit RPMs. This inherent stability allows the turbine to operate at high speeds without the need for extensive balancing mechanisms.

The high RPM capability of the Tesla turbine offers several advantages in practical applications. For instance, in power generation, high rotational speeds translate to increased shaft torque and power output. Similarly, in pump applications, such as pumping fluids in industrial processes or circulation systems, the high RPM enables efficient fluid movement with minimal energy loss.

Additionally, the Tesla turbine's high RPM capability makes it suitable for applications requiring precise control and

responsiveness, such as in propulsion systems for vehicles or aircraft, where rapid adjustments in power output are essential.

Overall, the Tesla turbine's ability to achieve high rotational speeds is a testament to its efficiency, versatility, and suitability for a wide range of applications where swift and reliable energy conversion is required.

II. Literature Review

Tesla Turbine has been a subject of CFD analysis before.Some of the works include: "The Tesla Turbine – A Comprehensive Review" provides a thorough examination of Tesla turbine design, performance, and applications, offering insights into its fluid mechanics and thermal characteristics[1] ."CFD Simulation of Tesla Turbines Performance Driven by Flue Gas of Internal Combustion Engine" investigates the performance of Tesla turbines using computational fluid dynamics (CFD) analysis, focusing on their efficiency when powered by flue gas from internal combustion engines [2]."CFD investigation on the performance analysis of Tesla turbine" employs CFD techniques to analyze the performance of Tesla turbines, contributing to the understanding of their operational characteristics and efficiency [3]."Design and Computational Analysis of 1 kW Tesla Turbine" presents the design and computational analysis of a 1 kW Tesla turbine, offering insights into its performance parameters and potential for practical applications [4].

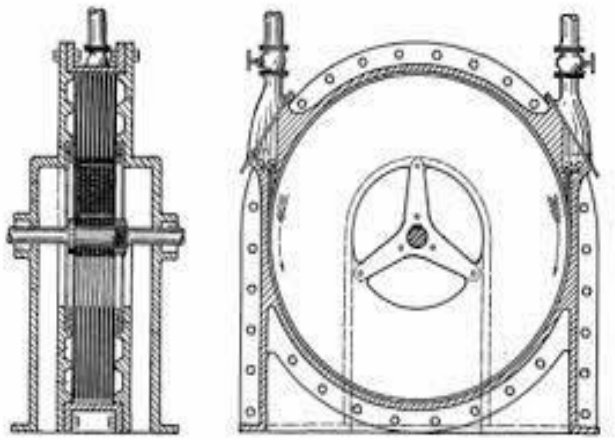


Figure 1. Original sketch of Tesla turbine [5]

Following are the design parameters which were studied extensively in literature:

1)Disc Thickness

Tesla's original turbine design featured discs with tapered edges to increase efficiency by reducing tip losses, which occur when fluid shifts from one side of the disc to the other, similar to airflow around airplane wingtips. Tesla used very thin discs to implement this idea, but they warped under high RPM stresses due to the limited materials and manufacturing techniques available. Li et al. conducted an experimental study using extremely thin discs (0.5 mm) to reduce shock losses, which occur when the working fluid impacts the disc edges [6]. They argued that the small disc thickness could effectively neglect shock losses. There are only a few studies on the effect of disc thickness, and further research is needed to determine the optimal thickness for different Tesla turbine applications.

2)Disc Gap

Krishnan et al. tested inter-disc gaps of 125 μm, 250 μm, and 500 μm across six rotor stacks, finding larger gaps increased RPM due to fewer discs and lighter stacks [10], while Romanin et al. concluded that smaller gaps improved efficiency by enhancing momentum transfer through increased viscous forces, supported by simulations and experiments [11]. Rice et al. suggested an optimal gap of twice the boundary layer thickness [7]. Borate et al. varied gaps from 0.5 mm to 2.5 mm using water, finding the smallest gap provided the best energy transfer due to overlapping boundary layers [8]. Breiter and Pohlhausen et al. analyzed fluid dynamics between rotating discs, stressing the importance of the disc gap and the Pohlhausen parameter in determining fluid rotation and flow profile, influencing turbine performance [12].

3) Number of discs

In a 1991 analytical conclusion, it was asserted that laminar flow can achieve rotor efficiency surpassing 95%, but at the cost of low fluid flow rates, attainable by increasing the number of discs. Studies, such as one by Romanin et al., tested rotors with varying disc counts, noting significant efficiency increases with more discs. Similarly, Hoya and Guha observed efficiency enhancements with increased disc numbers [13]. Tanet al. utilized computational fluid dynamics to optimize disc count, reducing head loss by decreasing discs from 21 to 13 [9]. This reduction in head loss occurs as fluid exits the disc gap, encountering a 90° turn towards the outlet, where flow separation and constriction to the vena contracta region occur.

4)RPM

Mohammad Zuber et al. from a computational model, it was found that higher RPM led to increased centrifugal forces and longer flow paths in the turbine [14]. While a maximum efficiency of 36.6% was achieved, the potential efficiency was estimated at 40%. Analyzing a simple model at different RPMs, it was observed that rotor speed affected streamline shape and fluid retention time, with longer paths improving momentum transfer and thus turbine efficiency. Numerical simulations at higher RPM showed 50% longer pathlines and increased efficiency. Experimental results generally showed a parabolic relationship between turbine efficiency and RPM, with torque

decreasing nonlinearly as RPM increases. Optimal RPM values were noted for specific pressure conditions, with efficiency peaking before declining. Inlet and outlet losses showed minimal variation with RPM, reaching minimum values at optimal rotor speeds. The turbine's maximum tested RPM was 25,000.

5) Power

Tesla's initial turbine utilized eight 6-inch diameter discs, generating 30 horsepower [5] . He later developed a turbine with 18-inch diameter discs capable of producing 200 horsepower with steam. While most turbines couldn't match Tesla's original design in power output, Li et al. observed a parabolic trend between output power and rotor speed, suggesting an optimal RPM for maximum power output. However, there's a lack of extensive study on the relationship between output power and other turbine parameters, indicating a need for further research in this area.

III. Construction

The Tesla turbine consists of multiple disks attached to a central shaft, enclosed within a metal casing. This design includes a nozzle that directs the incoming fluid tangentially to the disks, facilitating efficient flow. The fluid, in this case air, exits through an exhaust port located near the shaft's axis. As the fluid enters through the nozzle, it strikes the outer diameter of the disks and moves between them, following a centripetal path toward the center due to the rotation of the disks.

The disks are rotated by connecting them directly to a drive motor. However, achieving the high rotational speeds necessary for optimal performance poses challenges, such as the risk of disk warping or failure. To mitigate these issues, an external motor is connected to the central shaft of the Tesla turbine, enhancing its operational stability.

One notable advantage of the Tesla turbine is its versatility; it can also function as a Tesla pump when coupled with an external motor. Tesla turbines are particularly suited for applications that require lower electricity generation, commonly used in wastewater treatment plants, ventricular booster pumps, and within the petroleum industry.

In this project, we will explore the operating parameters of the Tesla turbine, such as disk diameter and rotation speed, and analyze their impact on turbine performance.

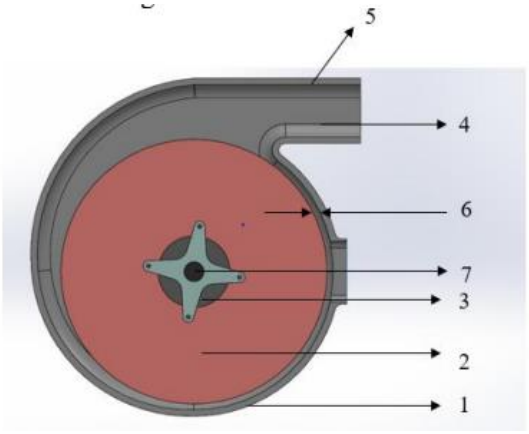


Figure 2. The Parts of Tesla turbine:1.casing,2.discs, 3.outlet port, 4. Inlet port, 5. Nozzle, 6.clearnce, 7.centre shaft

IV. Setting up

The objective of this project is to analyze how variations in disc diameter and rotating speed affect the pressure and velocity within a Tesla turbine. The modeling of the Tesla turbine was conducted using AutoCAD Fusion 360, while simulations were performed using SimScale. We investigated the impact of changes in both the disc diameter and the rotational speed, observing the resulting pressure and velocity contours in SimScale's solution fields. The radius of the disc was set to 90.076 mm.

Air is used as our primary fluid for simulation with its kinematic viscosity (v) and density (**ρ**) are set to 1.529e-5 m^2/s and $1.196 kg/m^3$.

The guage pressure was set to 0 Pa as it is assumed that the body is in a open , atmospheric condition during the analysis . The turbulent flow is given a value of $0.00375 m^2/s^2$, which signifies that the flow has very less turbulent characteristics , and is rather more laminar. Then the rotor was given a rotational velocity of 1200 rad/s or 11460 RPM for the first analysis and 600 rad/s or 5730 RPM for the second analysis . Then we proceeded by creating a mesh of our body .

Meshing is a key step in computational fluid dynamics (CFD) because it breaks down numerical calculations into smaller elements, leading to numerical approximations of fluid equations. The quality of the mesh affects the accuracy and reliability of CFD simulations because it determines how well the flow behaves in the captured domain. A properly created and optimized mesh ensures a reliable simulation by ensuring that important features such as boundary layers and vortices are correctly resolved. Meshing also increases efficiency by dividing the domain into manageable points, optimizing resources, and enabling simulation of real-life fluids.Following are the details about the obtained mesh :

1. Maximum Precision and Tolerances:

- Maximum precision of model and its entities: 1e-08 meters.
- Absolute small feature tolerance: 1e-05 meters.

2. Mesh Quality Metrics:
- tetEdgeRatio: This metric assesses the quality of tetrahedral elements. The values indicate the ratio of the longest edge to the shortest edge of tetrahedral elements. The average value is around 2.38.
- quadMaxAngle: This metric evaluates the maximum angle deviation within quadrilateral elements. The average maximum angle is around 96.79 degrees.
- triMaxAngle: This metric assesses the maximum angle deviation within triangular elements. The average maximum angle is around 83.10 degrees.
- triMinAngle: This metric evaluates the minimum angle within triangular elements. The average minimum angle is around 40.78 degrees.
- volumeRatio: This metric assesses the ratio of volume between tetrahedral elements. The average ratio is around 1.56.
- tetAspectRatio: This metric evaluates the aspect ratio of tetrahedral elements. The average aspect ratio is around 2.39.
- nonOrthogonality: This metric evaluates the non-orthogonality of the mesh. The average non-orthogonality angle is around 25.48 degrees.
- skewness: This metric assesses the skewness of the mesh elements. The average skewness is around 0.15.
- aspectRatio: This metric evaluates the aspect ratio of mesh elements. The average aspect ratio is around 1.99.

3. Overall Mesh Quality:
- The overall mesh quality, based on the 99-percentile of the quality metrics, is approximately 0.269171. This overall quality is computed from non-orthogonality, skewness, and aspect ratio, each with their respective weights.

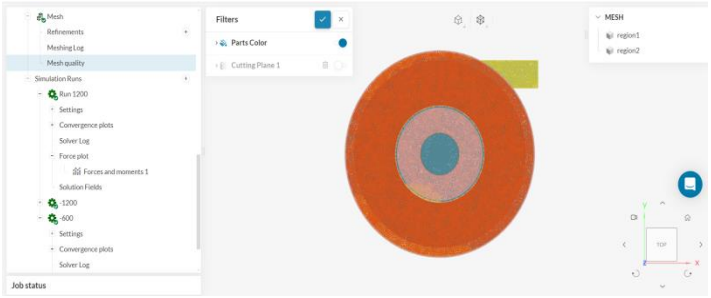


Figure 3. Meshing

V. Results

1]Outputs for the Turbulent kinematic viscosity , pressure and velocity contours respectively for analysis done at 1200 rad/s or 11460 RPM :

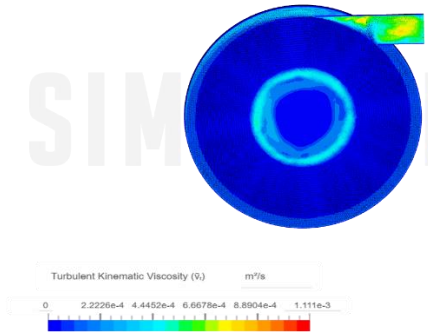


Figure 4. Turbulent kinematic viscosity

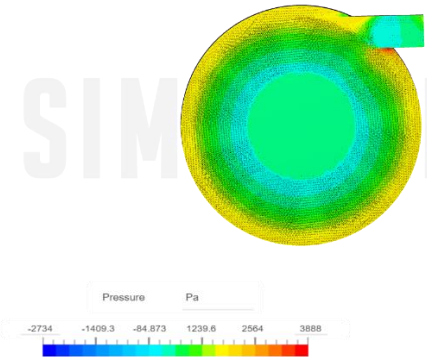


Figure 5. Pressure

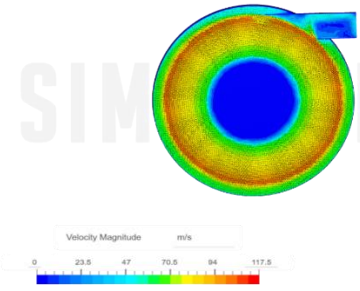


Figure 6. Velocity

2] Outputs for the Turbulent kinematic viscosity , pressure and velocity contours respectively for analysis done at600 rad/s or 5730 RPM

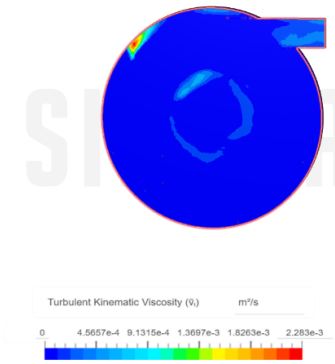


Figure 7. Turbulent kinematic viscosity

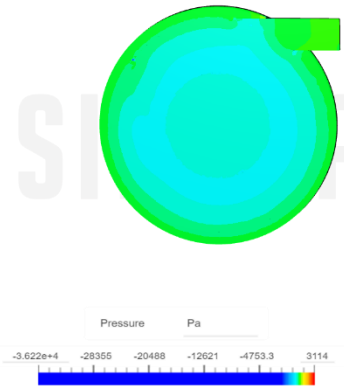


Figure 8. Pressure

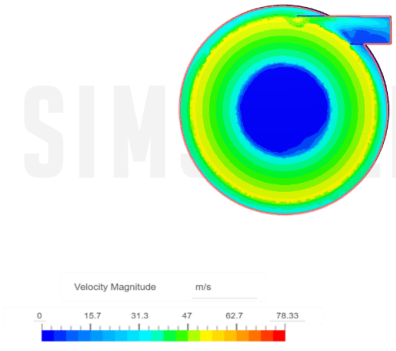


Figure 9. Velocity

VI. Calculation

To find the torque output of the turbine for both cases , which was 1200 rad/s and 600 rad/s ,

Input Values:

For t = 60 sec, ω = 1200 rad/s:

Pressure Force (x, y, z): 0.01088958459614, 0.004497649641305, -0.0002215724024116

Pressure Moment (x, y, z): -0.000780311391932, -0.000435855397879, -0.000004325961244604

Viscous Force (x, y, z): 0.01334247552991, -0.002506763752197, 0.00003962831523194

Viscous Moment (x, y, z): 0.000004091348397089, 0.000006236787679175, -0.1201584558067

Porous Force (x, y, z): 0, 0, 0

For t = 60 sec, ω = 600 rad/s:

Pressure Force (x, y, z): 0.014079482, 0.008765205, 0.008550792

Pressure Moment (x, y, z): -0.000733943, -0.000414452, 5.13E-07

Viscous Force (x, y, z): 0.001073359, 0.00674697, 0.000792854

Viscous Moment (x, y, z): 3.72E-06, -5.86E-07, 0.039148081

Porous Force (x, y, z): 0, 0, 0

Total Torque Output:

At t = 60 sec, ω = 1200 rad/s:

Total Torque_x ≈ -0.0007762 Nm

Total Torque_y ≈ -0.0004296 Nm

Total Torque_z ≈ -0.1202 Nm

Magnitude of Torque ≈ 0.1234 Nm

At t = 60 sec, ω = 600 rad/s:

Total Torque_x ≈ 0.000339416 Nm

Total Torque_y ≈ 0.006332518 Nm

Total Torque_z ≈ 0.039148594 Nm

Magnitude of Torque ≈ 0.04106 Nm

Comparisons:

Comparing the torques at t = 60 sec for ω = 1200 rad/s and ω = 600 rad/s, we observe:

The magnitude of the torque at 1200 rad/s is approximately 0.1234 Nm, while at 600 rad/s, it is approximately 0.04106 Nm.

The power produced at 1200 rad/s is approximately 0.148KW, while at 600 rad/s, it is approximately 0.0246KW.

Observations:

The torque output is highly dependent on the rotational speed, with higher speeds resulting in larger torques, and lower speeds resulting in smaller torques.

The direction of torque also changes with the rotational speed, indicating different rotational behaviours of the turbine.

VII. 3D Printed Prototype

In our study we also created a demonstrative prototype of tesla turbine using additive manufacturing technique. In much research we found the theoretical and computational study of tesla turbine, there was less work done in manufacturing of tesla turbine. So, in our project we have created a model which can demonstrate how a tesla turbine works. The simulation and analysis which we have carried out is not directly connected to the model or the design parameters, materials are not same. We tested the model using compressed air and got the results.

For the fabrication of our Tesla turbine model, we utilized 3D printing technology, which allowed us to create intricate and precise components. The chosen 3D printer was a Creality Ender 3, noted for its reliability and cost-effectiveness. We used PLA (Polylactic Acid) as the printing material due to its ease of use, minimal warping, and good surface finish.



Figure 10. 3d printed Tesla Turbine

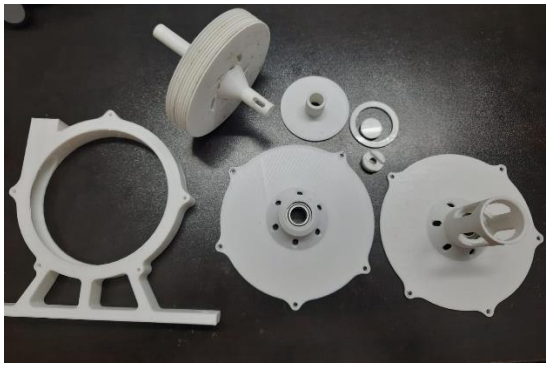


Figure 21. Parts of Tesla Turbine

Parameters of 3d printing

The 3D printing process was meticulously configured with the following parameters:

- **Nozzle Size:** 0.4mm
- **Nozzle Temperature:** 205°C
- **Bed Temperature:** 50°C
- **Layer Height:** 0.18mm

These settings were selected to achieve optimal print quality and performance for the turbine model. The nozzle temperature of 205°C ensured proper extrusion and strong layer adhesion, while the bed temperature of 50°C provided stability and prevented warping. The 0.4mm nozzle allowed for fine detail resolution, and the 0.18mm layer height balanced detail with print time.

Using these parameters, we achieved a high-quality printed model that met our design specifications and functional requirements, demonstrating the effectiveness of 3D printing in prototyping and manufacturing complex components such as the Tesla turbine.

VIII. Observations

After thorough comparison of the Turbulent kinematic viscosity, pressure and velocity contours at 11460 and 5730 , it is evident that as the rpm of the disc increases , more mechanical stress is induced onto the disc , which may cause disc failure . This happens to be tesla turbines major setback as it heavily relies on high rpm to increase its output and efficiency.

The high RPM (revolutions per minute) and disc wear problem is one of the key challenges associated with Tesla turbines. Here's a breakdown of why this is a concern:

1. High RPM: Tesla turbines typically operate at very high speeds compared to conventional turbines. The discs within the turbine can rotate at speeds exceeding tens of thousands of RPM. While

this high speed can contribute to the turbine's compactness and efficiency in theory, it also poses practical challenges.

2. Centrifugal Forces: At such high RPM, the centrifugal forces acting on the rotating discs are substantial. These forces can exert significant stress on the material of the discs, potentially leading to deformation or failure if not properly addressed.

3. Friction and Wear: The high-speed rotation of the discs also generates considerable friction within the turbine. This friction can cause wear and tear on the surfaces of the discs, leading to degradation over time. The wear becomes more pronounced as the RPM increases.

4. Material Considerations Finding materials that can withstand the stresses induced by the high RPM of Tesla turbines is crucial. The discs need to be made from materials with high strength, durability, and resistance to wear. However, such materials can be expensive or challenging to manufacture, adding to the overall cost and complexity of the turbine.

5. Maintenance Requirements: Due to the wear caused by high RPM operation, Tesla turbines may require more frequent maintenance to replace worn components and ensure continued operation. This can increase operational costs and downtime, reducing the overall viability of the turbine for certain applications.

Addressing the high RPM and disc wear problem is essential for improving the reliability, efficiency, and longevity of Tesla turbines. Research efforts focus on developing advanced materials, optimizing turbine designs, and implementing effective lubrication and cooling systems to mitigate these challenges. By overcoming these obstacles, Tesla turbines could become more competitive alternatives in various applications, especially where their unique advantages, such as simplicity and flexibility, outweigh the drawbacks.

IX. Conclusion

With growing need for renewable alternatives for enegy necessitating use of newer techniques particularly in field of hydro-power energies. The use of the Tesla turbine in today's scenario is crucial for meeting these demands, requiring substantial enhancements in its performance. Compared to traditional turbines, the Tesla turbine offers greater economic viability. An improved design holds promise for the future, potentially replacing other turbines that contribute to water resource contamination and environmental pollution. Selecting the appropriate design is pivotal for achieving superior turbine performance and integrating it into hydro-power applications to surpass conventional turbine technology. This project focuses on augmenting pressure and velocity within the Tesla turbine to elevate its efficiency and power output. Through this endeavor, it was observed that increasing the disc rotating speed also led to enhanced power output. A graph depicting power output versus rpm illustrates that power output peaks at higher rpm levels, indicating a crucial parameter for Tesla turbine design optimization. Moreover, parameters related to nozzles, such as nozzle inclination, can be further optimized to refine the working of the Tesla turbine.

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